Feedback

Workshop 7

Workshop 7 feedback

- In the seventh workshop we looked at the parallel performance of a program which calculates the Mandelbrot set
- The first step in the performance test was to measure the serial run time
- The second step was to run the OpenMP version of the program with different numbers of threads and plot the speed-up against number of threads
- The third step was to test different loop schedules and see which one performs best

Workshop 7

Question 1: What is the value of T_0 for this version of the program?

The serial run time give us T_0:

```
Starting serial calculation

real	1m30.985s
user	1m30.901s
sys	0m0.085s
slurm-498590.out (END)
```

$$T_0 = 90.985\text{s}$$

Workshop 7

Scaling test (default schedule):

```
Starting calculation on 2 threads

real	1m2.143s
user	1m30.902s
sys	0m0.074s
Starting calculation on 4 threads

real	0m53.233s
user	1m31.190s
sys	0m0.079s
Starting calculation on 8 threads

real	0m27.231s
user	1m33.808s
sys	0m0.073s
Starting calculation on 12 threads

real	0m21.716s
user	1m37.048s
sys	0m0.073s
Starting calculation on 16 threads

real	0m15.757s
user	1m40.058s
sys	0m0.069s
slurm-498631.out (END)
```

No. of Threads	Time (s)
2	62.143
4	53.233
8	27.231
12	21.716
16	15.757

Workshop 7

Scaling test (default schedule):

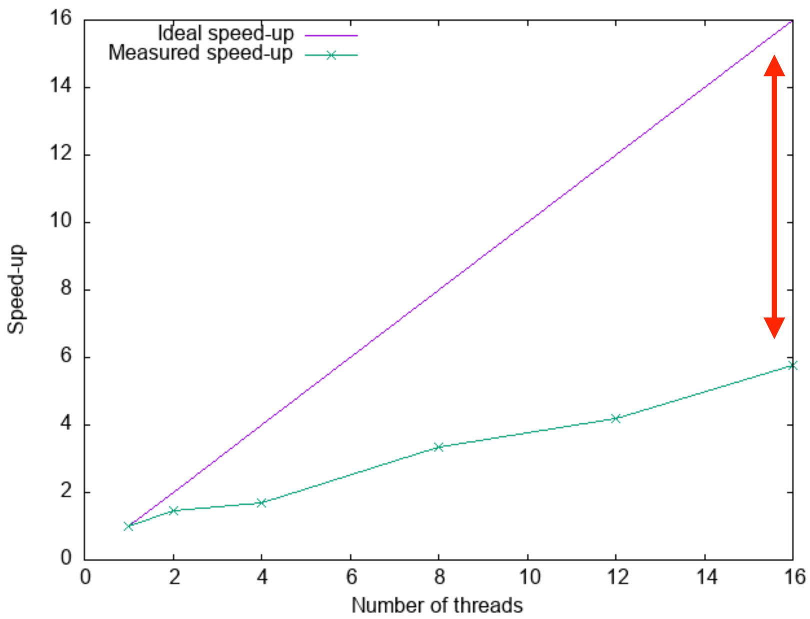
No. of Threads	Time (s)	Speed-up	Efficiency
1 (serial)	90.985	1	1
2	62.143	1.464	0.732
4	53.233	1.709	0.427
8	27.231	3.341	0.418
12	21.716	4.190	0.349
16	15.757	5.774	0.361

$$S_N = \frac{T_N}{T_0}$$

Workshop 7

- Plot the parallel speed-up (vertical axis) against the number of threads (horizontal axis) and compare this with the ideal speed-up.

  Question 2: Is the program making efficient use of the available resources?



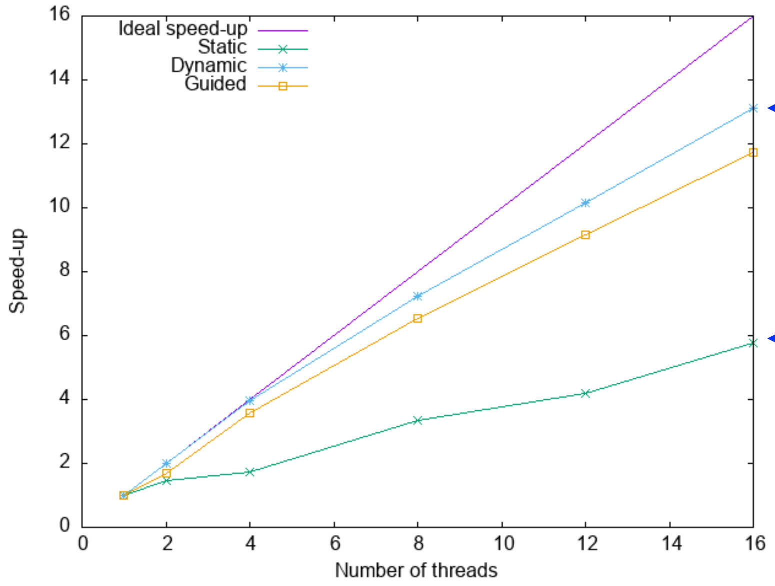
Big difference between measured and ideal speed-up. Not making efficient use of the available resources

Workshop 7

- Add a clause to the `#omp parallel for` which specifies a loop schedule and re-run the scaling test using static, dynamic and guided schedules. Remember to re-compile the executable after changing the schedule.

  Question 3: Which loop schedule performs best and why?

  Question 4: What is the default loop schedule?



Dynamic performs best because it load balances at run time with most flexibility

Default schedule is static

Workshop 7 summary

- Workshop 7 was about strong scaling a calculation of the Mandelbrot set
- The workload is poorly balanced due to an inner loop with a variable number of iterations
- Using a dynamic schedule greatly improves the performance